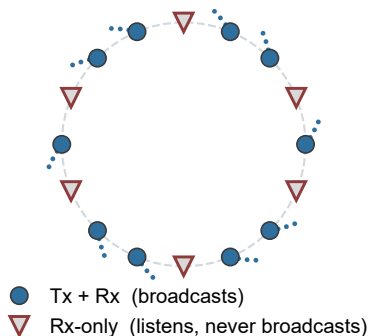
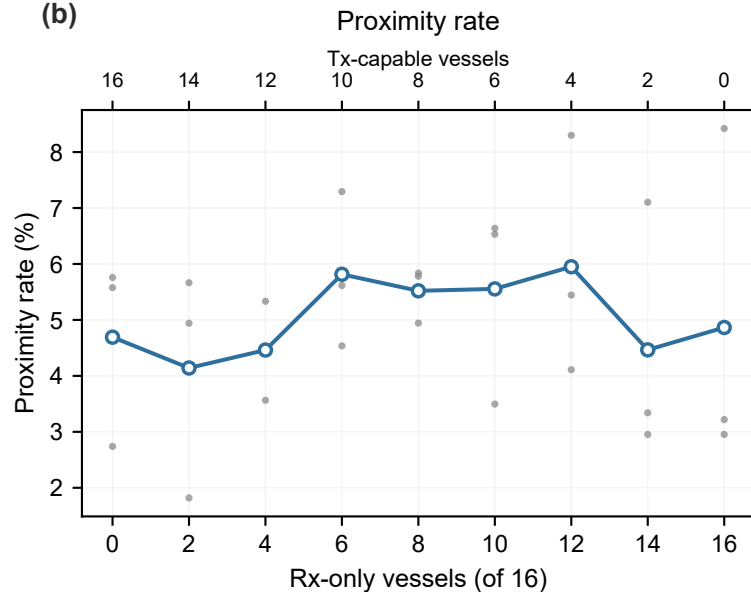


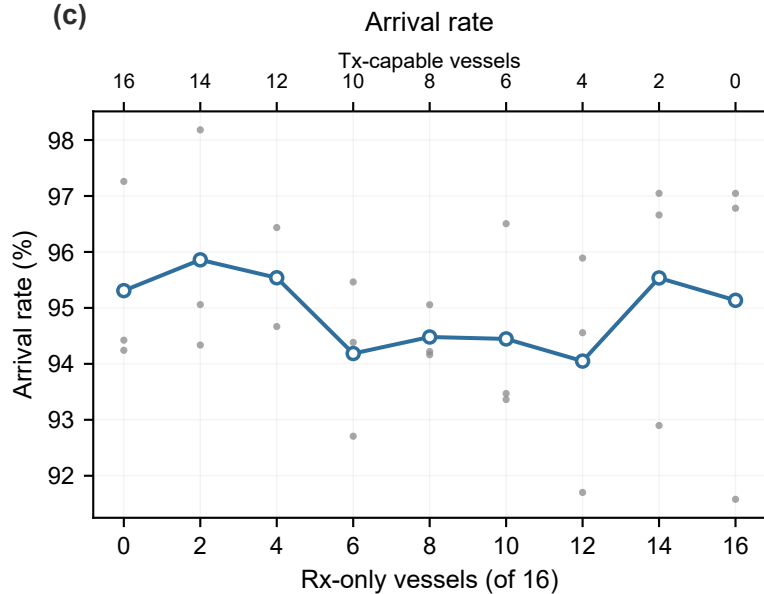
**(a)** Fleet composition  
 Spawn ring — 16 vessels, 6 receive-only



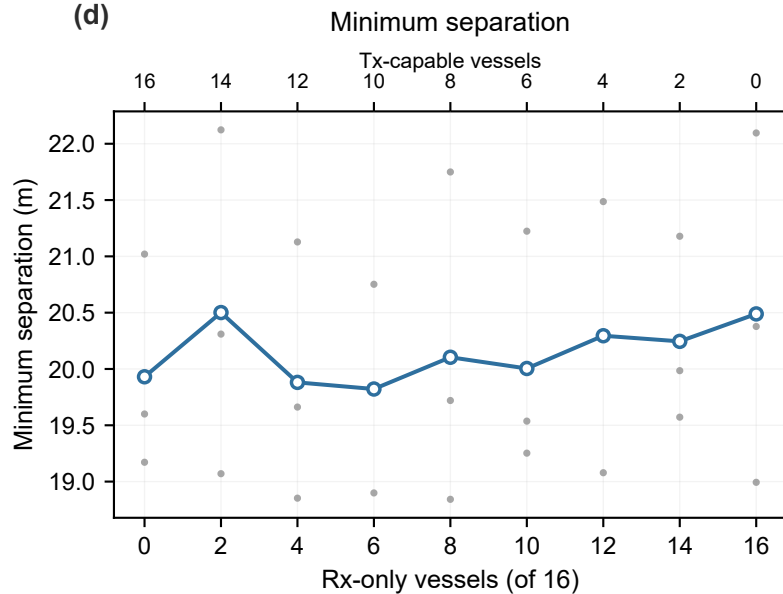
**(b)**



**(c)**



**(d)**



v12mix hub, mute-TX (rx-only) sweep, 3 seeds, sample-weighted fleet.

Each  $n_{rx}$  is an isolated eval (64 envs) matching Fig1 budget —  $n_{rx}=0$  anchors to Fig1 ON (~95%).

Claim: cutting transmitters does not collapse the fleet while receivers still hear.  $n_{rx}=16$  is test-time message ablation on the ON policy (not Fig1 OFF-trained).

Step-legacy COLREGs panel removed (was ~63% and not PRIMARY).